AUTOMATIC TRANSAXLE REMOVAL/INSTALLATION [EW6A-EL]

id0517l2117200

Procedure Before i-ELOOP-Related Part Servicing

Warning

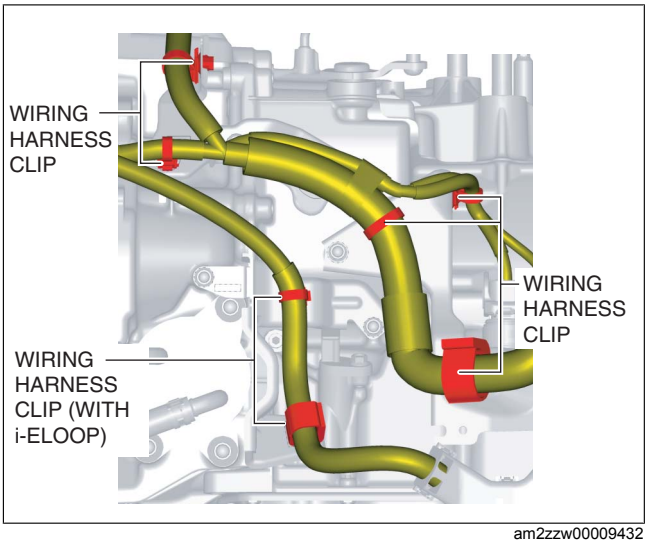
- **If the terminal of one of the i-ELOOP-related parts contacts the vehicle body with the service plug of the capacitor connected, it could cause a fire and damage to electrical parts. In addition, if the terminal contacts your body, it could cause electrocution. Disconnect the service plug of the capacitor before disconnecting the i-ELOOP-related part terminal. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)**
  - **— Battery cable terminal of capacitor (i-ELOOP)**
  - **— Engine wiring harness terminal of capacitor (i-ELOOP)**
  - **— Battery cable terminal of DC-DC converter (i-ELOOP)**
  - **— Generator terminal B cable**

Caution

- **If the automatic transaxle has been replaced, perform TCM configuration. (See TCM CONFIGURATION [EW6A-EL].)**
- **If the automatic transaxle has been replaced, perform initial learning. (See INITIAL LEARNING [EW6A-EL].)**
- **Performing the following procedures without first removing the front ABS wheel-speed sensor may possibly cause an open circuit in the harness if it is pulled by mistake. Before performing the following procedures, disconnect the front ABS wheel-speed sensor (axle side) and fix it to an appropriate place where the sensor will not be pulled by mistake while servicing the vehicle.**
- **Secure the steering wheel using tape or a cable to prevent the steering shaft from rotating after disconnecting the steering shaft. If the steering wheel rotates after the steering shaft and the steering gear and linkage are disconnected, the internal parts of the clock spring could be damaged.**

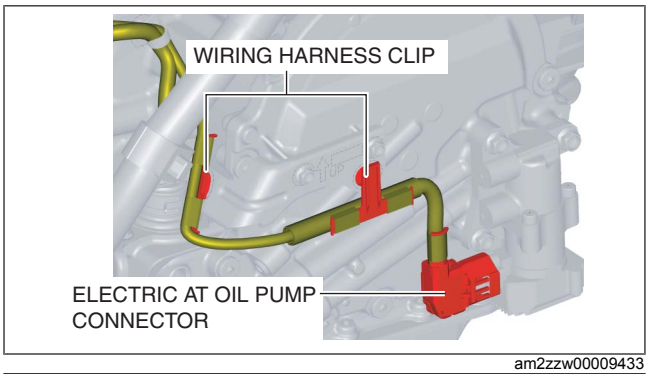
Removal

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the splash shield (LH). (See SPLASH SHIELD REMOVAL/INSTALLATION.)
3. Disconnect the service plug. (With i-ELOOP) (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
4. Remove the air cleaner, air hose and fresh air duct as a single unit. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Remove the battery and battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Remove the DC-DC converter (i-ELOOP). (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
7. Remove the turbocharger air outlet pipe. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8. Disconnect the wiring harness clip from the transaxle.



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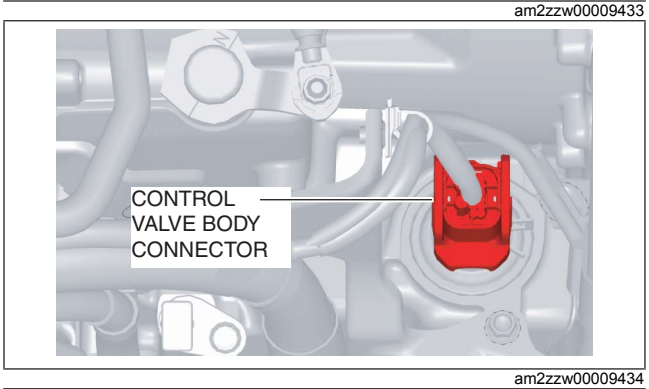
9. Disconnect the electric AT oil pump connector and wiring harness clip from the transaxle.



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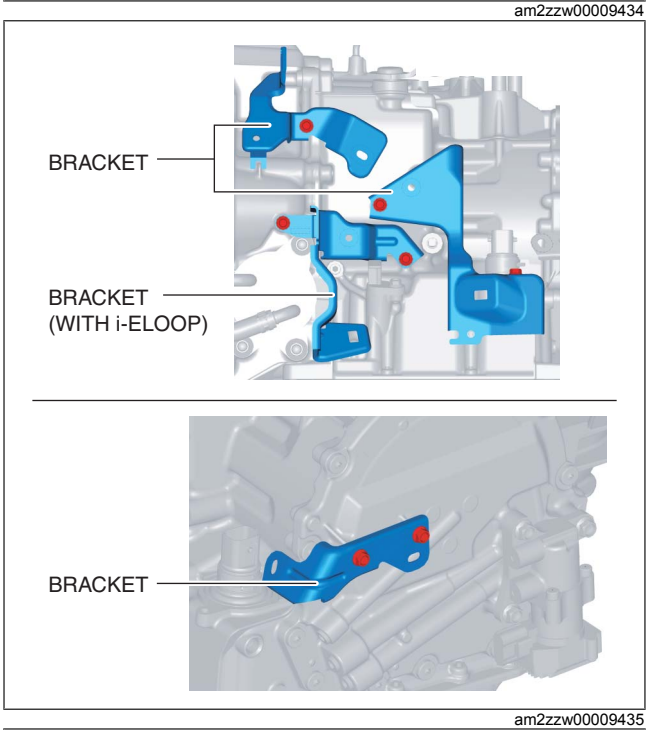
10. Disconnect the control valve body connector.

Caution
- **Make sure that your hand does not touch the terminal as the connector terminal could be damaged.**
- **Water or foreign objects entering the connector can cause a poor connection or corrosion. Be sure not to drop water or foreign objects on the connector when disconnecting it.**



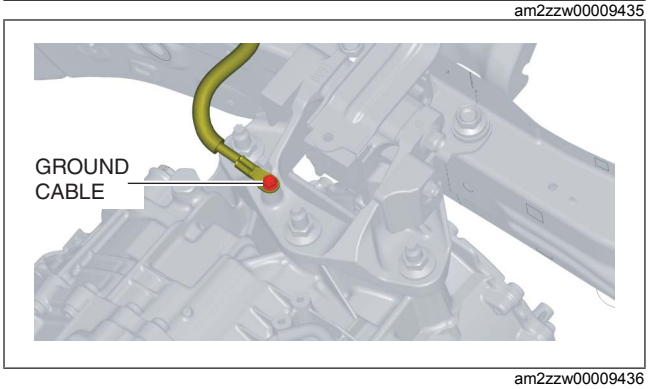
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11. Remove the bracket from the transaxle.



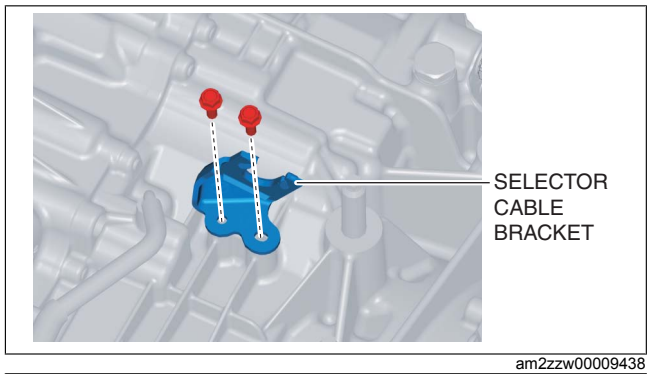
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12. Disconnect the ground cable.
13. Disconnect the clip and selector cable from the transaxle. (See AUTOMATIC TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION.)



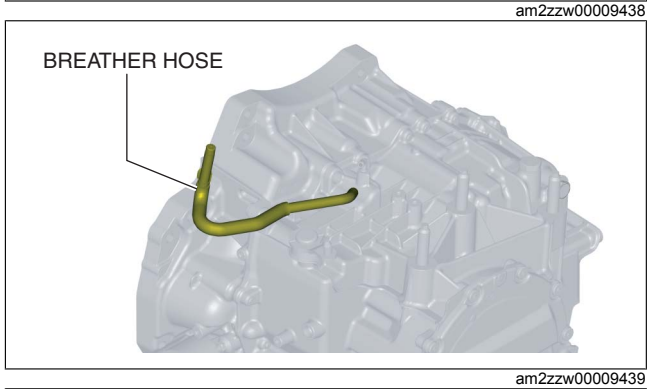
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14. Remove the selector cable bracket.



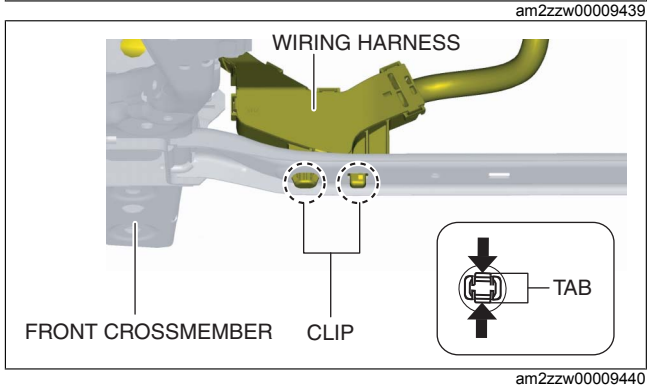
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15. Disconnect the breather hose from the transaxle.
16. Remove the joint cover. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
17. Disconnect the intermediate shaft from the steering gear and linkage. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
18. Remove the front tires. (See GENERAL PROCEDURES (SUSPENSION).)
19. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/ INSTALLATION.)

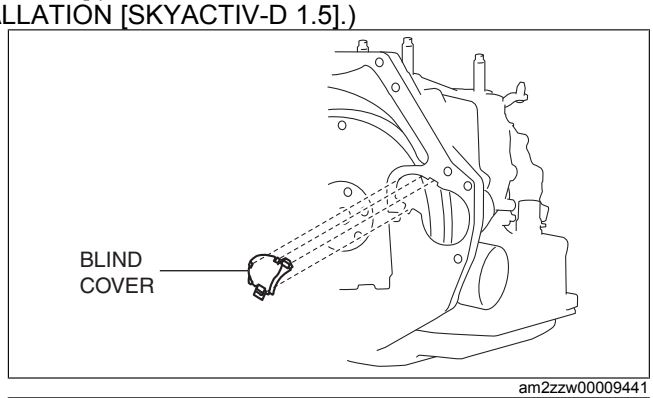


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20. Disconnect the wiring harness from the front crossmember. (With i-ELOOP)
21. Remove the front under cover No.1. (See FRONT UNDER COVER No.1 REMOVAL/ INSTALLATION.)
22. Drain the engine coolant and water-cooled charge air cooler coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].) (See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
23. Drain the ATF. (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [EW6A-EL].)
24. Remove the electric water pump. (See ELECTRIC WATER PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
25. Remove the starter. (See STARTER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
26. Remove the blind cover.

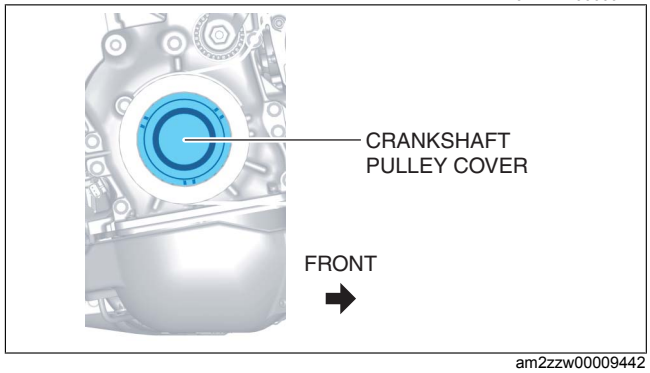


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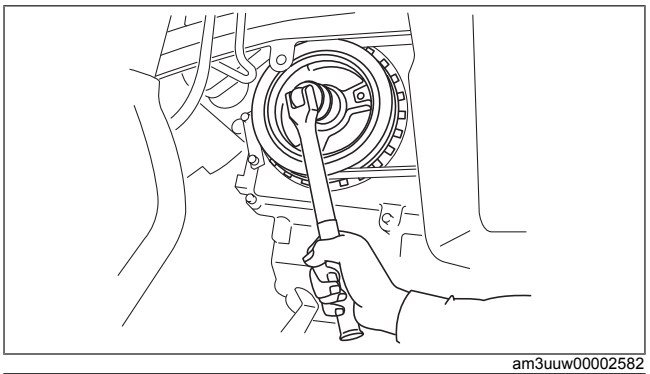
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27. Remove the crankshaft pulley cover.



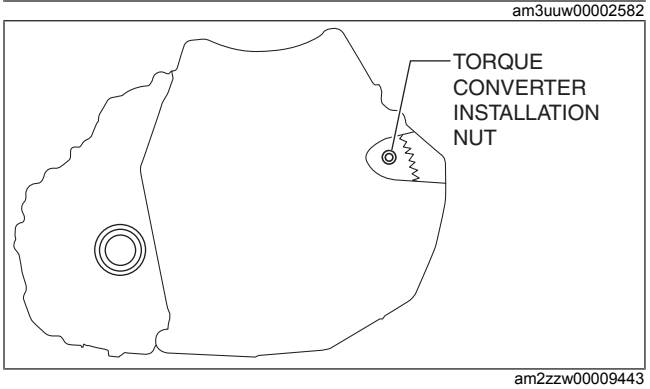
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28. Hold the crankshaft pulley to prevent torque converter from rotating.



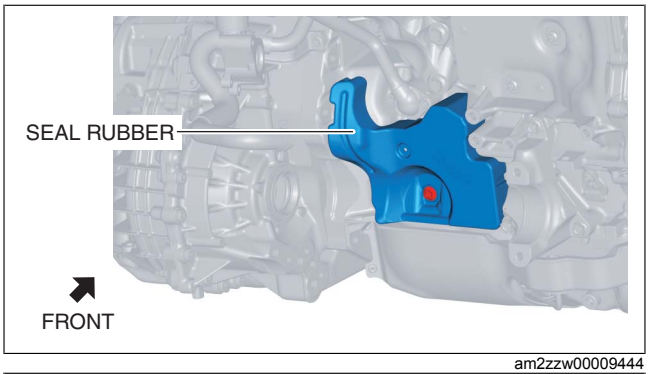
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29. Remove the torque converter installation nuts from the starter installation hole.
30. Disconnect the front ABS wheel-speed sensors from the steering knuckles. (See FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION.)
31. Disconnect the clips securing the brake hose from the front shock absorbers. (See BRAKE HOSE (FRONT) REMOVAL/INSTALLATION.)
32. Disconnect the tie-rod end ball joints from the steering knuckles. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



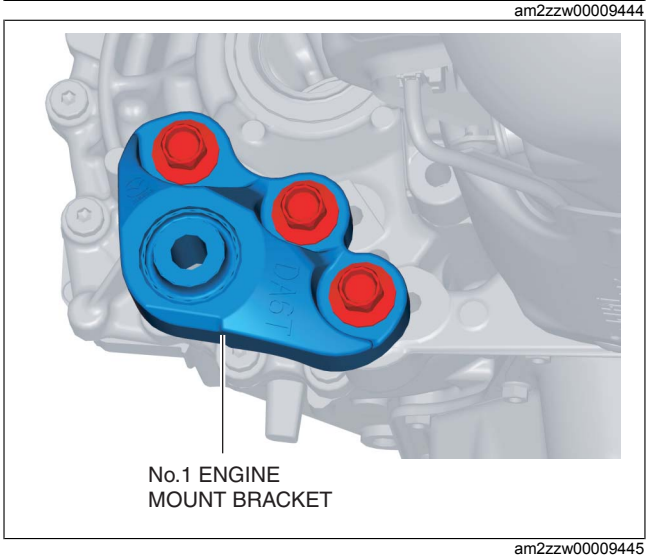
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33. Disconnect the front lower arms from the steering knuckles. (See FRONT LOWER ARM REMOVAL/INSTALLATION.)
34. Disconnect the front stabilizer control links from the front stabilizer. (See FRONT STABILIZER REMOVAL.)
35. Disconnect the front drive shaft (LH) from the transaxle. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
36. Disconnect the front drive shaft (RH) from the transaxle. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
37. Remove the seal rubber.
38. Remove the insulator. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
39. Remove the front crossmember component and No. 1 engine mount rubber as a single unit. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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40. Remove the No.1 engine mount bracket from the transaxle.

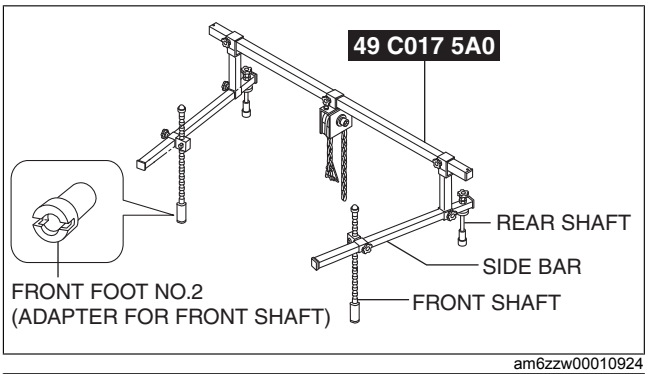


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41. Install the **SST** (49 C017 5A0, 49 UN30 3050, 49 L017 5A0) using the following procedures.

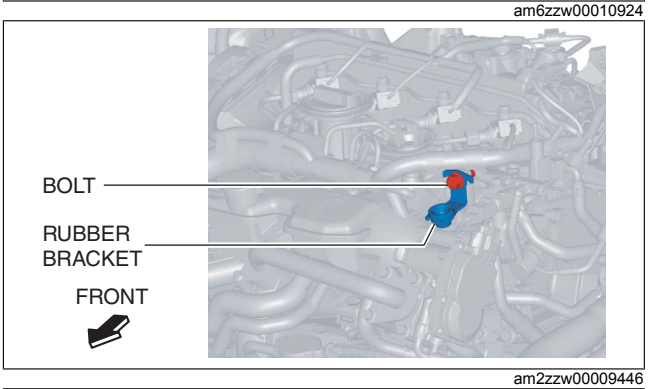
Caution
- **Refer to the SST (49 C017 5A0) instruction manual for the basic handing procedure.**

(1) Install one front foot No.2 to each of the left and right front shafts of the **SST** (49 C017 5A0).



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(2) Remove the rubber bracket.



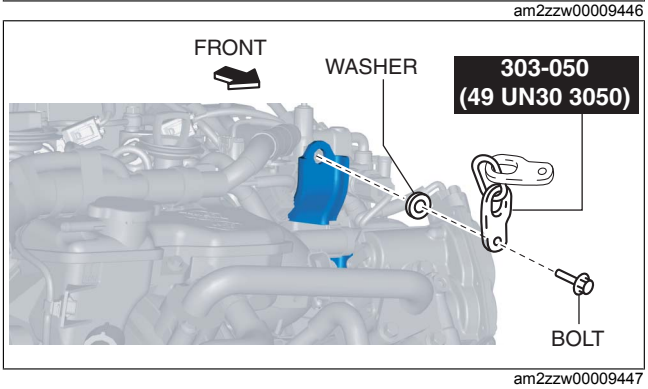
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(3) Install the **SST** (49 UN30 3050) to the position shown in the figure using the following bolt and washer.

Bolt
Part number 99794 1025 or an M10—1.25, length 25 mm {0.98 in}

Washer
approx. 3 mm {0.1 in} thickness

Tightening torque
38—52 N·m {3.9—5.3 kgf·m, 29—38 ft·lbf}



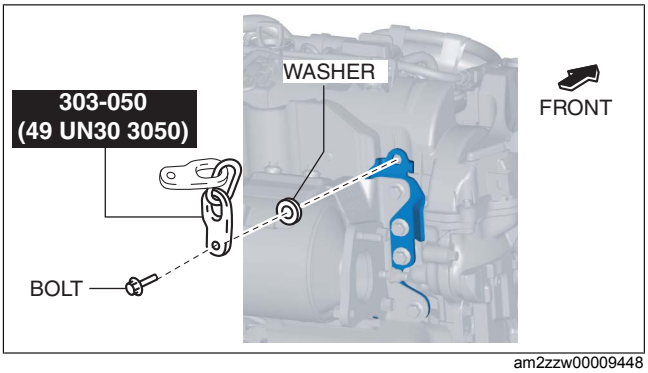
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(4) Install the **SST** (49 UN30 3050) to the position shown in the figure using the following bolt and washer.

Bolt
Part number 99794 1025 or an M10—1.25, length 25 mm {0.98 in}

Washer
approx. 3 mm {0.1 in} thickness

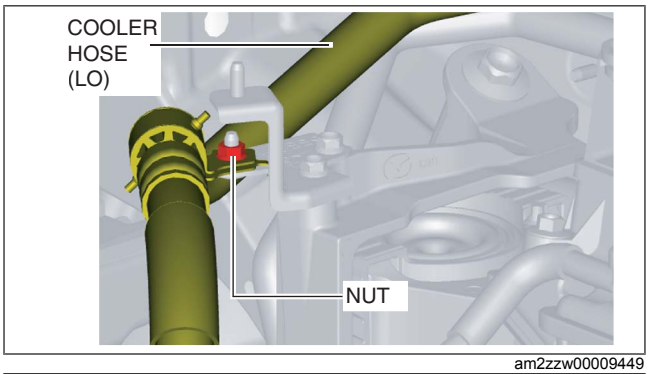
Tightening torque
38—52 N·m {3.9—5.3 kgf·m, 29—38 ft·lbf}



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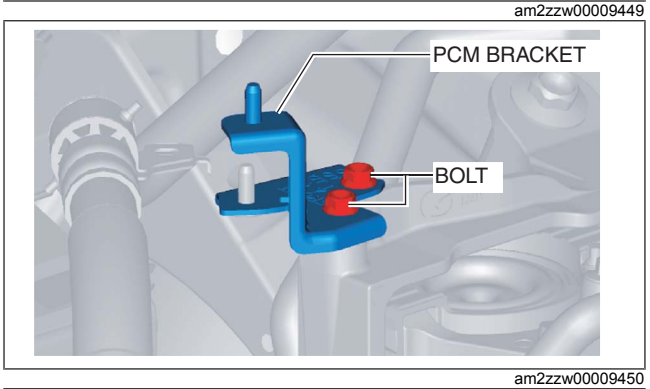
(5) Remove the PCM component with the PCM connector remaining connected, and set the PCM component aside. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

(6) Remove the nut shown in the figure and set the cooler hose (LO) aside.



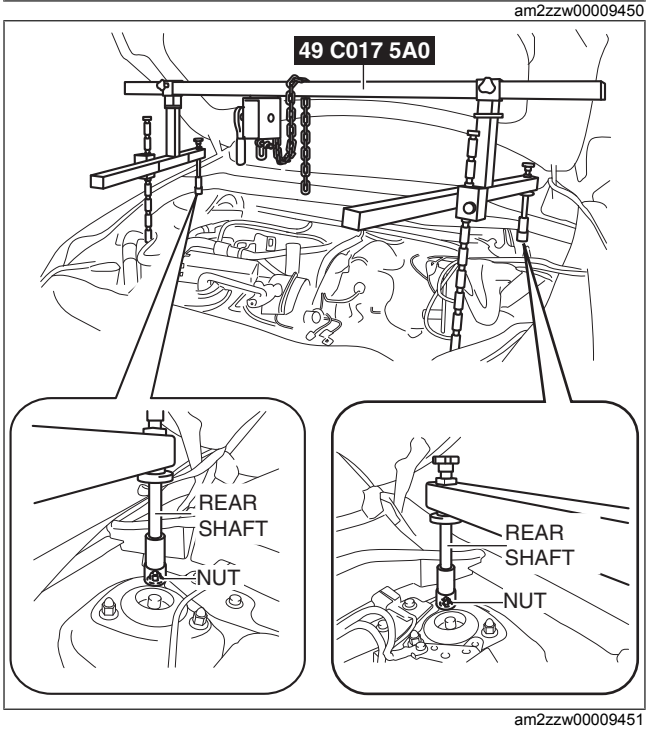
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(7) Remove the PCM bracket.



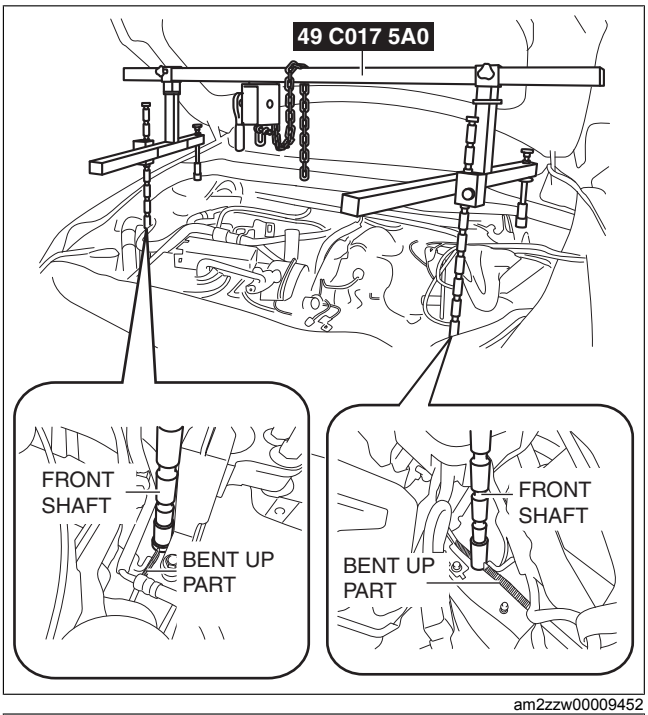
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(8) Set the rear shafts of the **SST** (49 C017 5A0) on the left/right front shock absorber nuts (top) as shown in the figure.



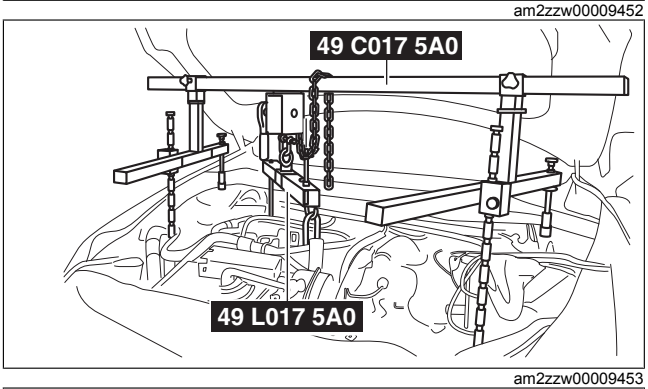
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(9) Set the front shafts of the **SST** (49 C017 5A0) on the bent up part of the frame as shown in the figure.



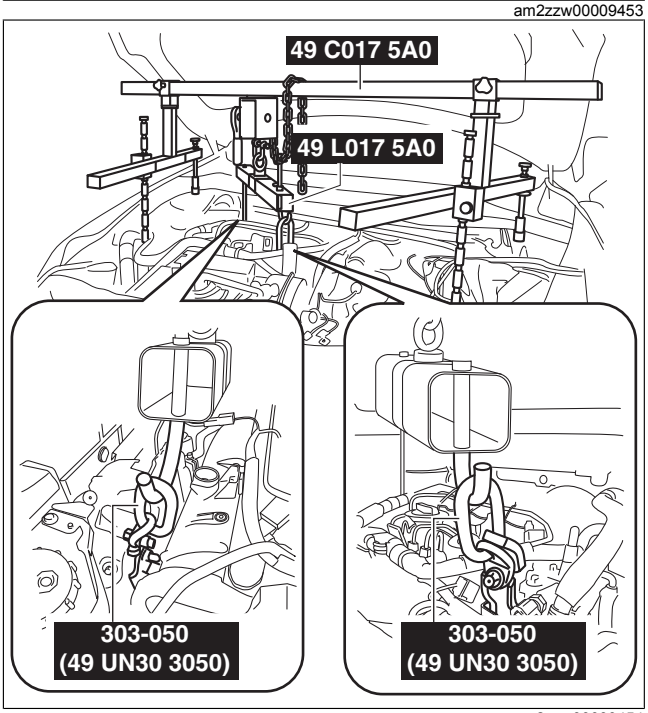
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(10) Install the **SST** (49 L017 5A0) to the **SST** (49 C017 5A0) as shown in the figure.



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(11) Install the **SST** (49 L017 5A0) to the **SST** (49 UN30 3050) with the hook of the **SST** (49 L017 5A0) facing outward.
(12) Adjust the height of the left and right side bars so that they are leveled, then tighten each part of the **SST** (49 C017 5A0).
(13) Apply tension to the chain to secure the engine.



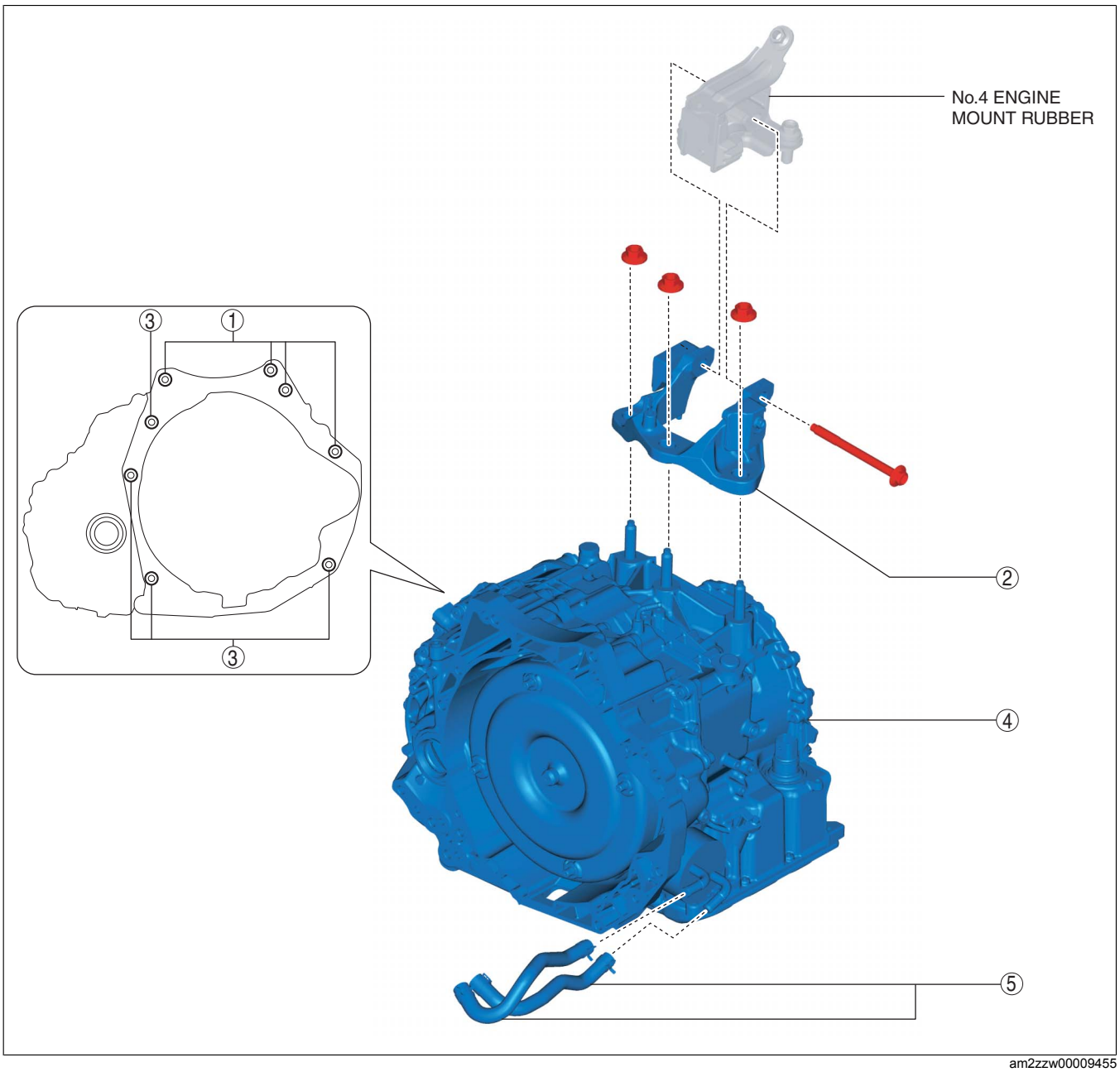
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42. Remove in the order shown in the table.

Warning
- **Improperly jacking a transaxle is dangerous. It can slip off the jack and may cause serious injury.**

Caution

- **To prevent the torque converter and transaxle from separating, remove the transaxle without tilting it toward the torque converter.**



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1	Transaxle mounting bolts (upper side)
2	No.4 engine mount bracket (See No.4 engine mount bracket removal note.)
3	Transaxle mounting bolts (lower side) (See Transaxle mounting bolt (lower side) removal note.)
4	Transaxle (See No.4 engine mount bracket removal note.) (See Transaxle mounting bolt (lower side) removal note.)
5	Water hose (Automatic transaxle replacement) (See OIL COOLER REMOVAL/INSTALLATION [EW6A-EL].)

No.4 engine mount bracket removal note

Caution

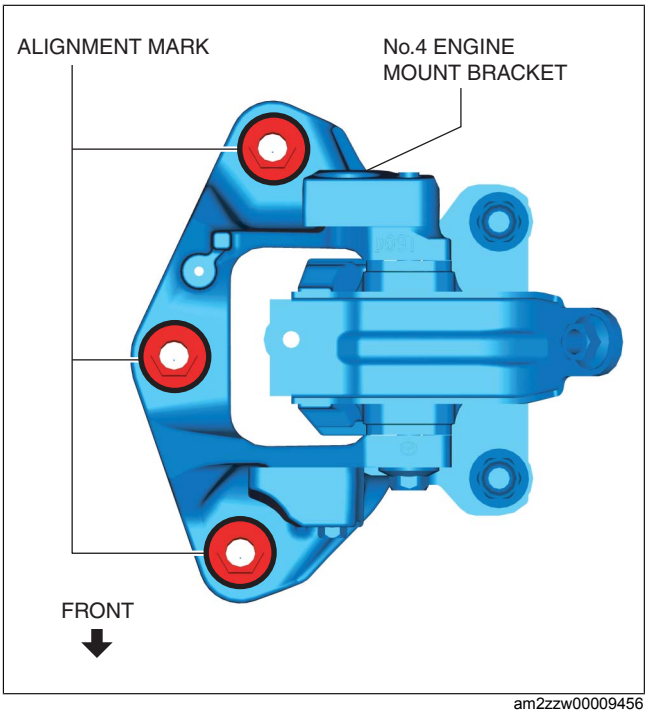
- **A slot has been adopted for the No.4 engine mount bracket installation hole. If the No.4 engine mount bracket is deviated from the original position when installing the No.4 engine mount, engine noise or vibration could increase. When removing the No.4 engine mount, place alignment marks on the No.4 engine mount bracket so that they can be assembled to the same positions as before removal.**

1. Place alignment marks on the locations shown in the figure so that they can be assembled to the same positions as before removal.

   Note

   - Paint so that the outline of the nut is framed on the bracket side.

2. Remove the No.4 engine mount bracket.



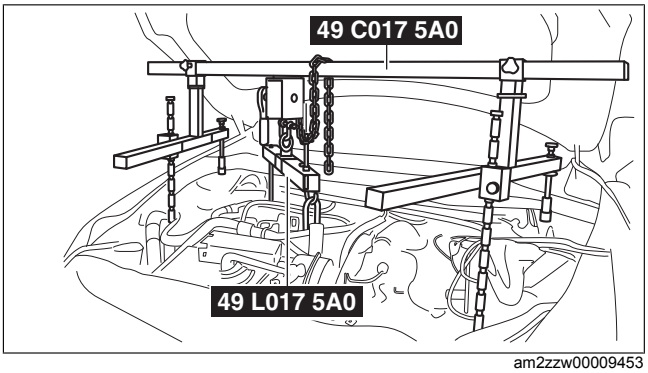
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Transaxle mounting bolt (lower side) removal note

Warning

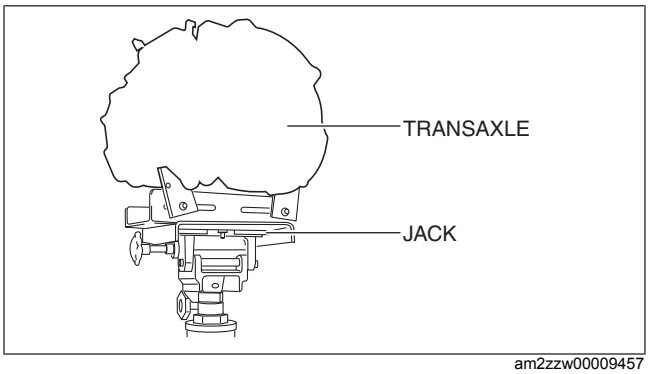
- **Remove the transaxle carefully, holding it steady. If the transaxle falls it could be damaged or cause injury.**

1. Adjust the **SST** (49 C017 5A0) and lean the engine toward the transaxle.



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2. Support the transaxle on a jack.
3. Remove the transaxle mounting bolts (lower side).
4. Remove the transaxle.

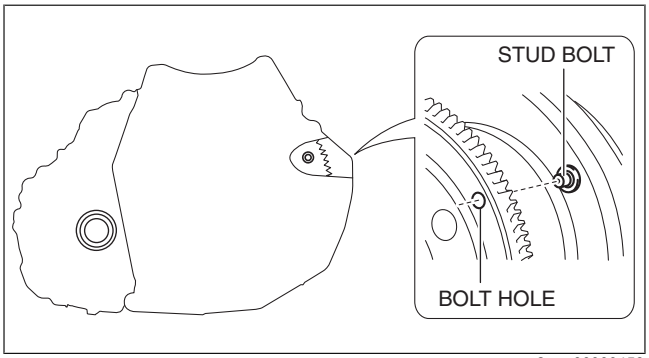


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Installation

1. Install the water hose. (Automatic Transaxle Replacement) (See OIL COOLER REMOVAL/INSTALLATION [EW6A-EL].)
2. Verify that the torque converter stud bolts are inserted into the drive plate bolt holes from the starter installation hole.

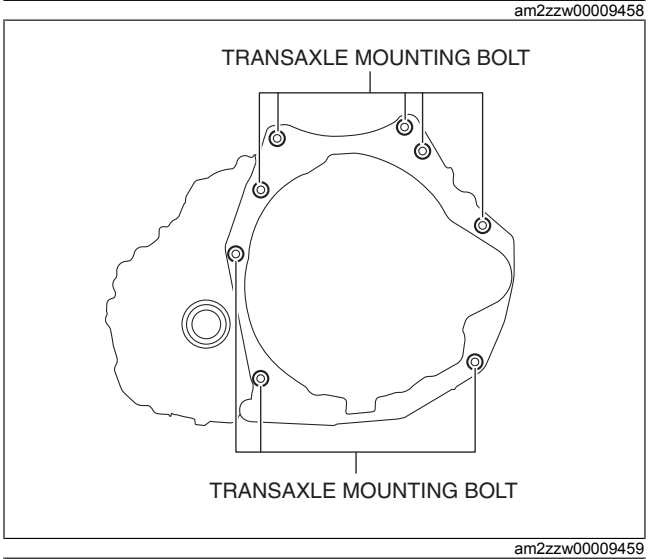
   Caution
   - **Verify that the torque converter stud bolts are inserted into the drive plate bolt holes, and install the transaxle installation bolts. Otherwise, the drive plate could be damaged.**



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3. Install the transaxle mounting bolts.

   Tightening torque
   38—52 N·m {3.9—5.3 kgf·m, 29—38 ft·lbf}

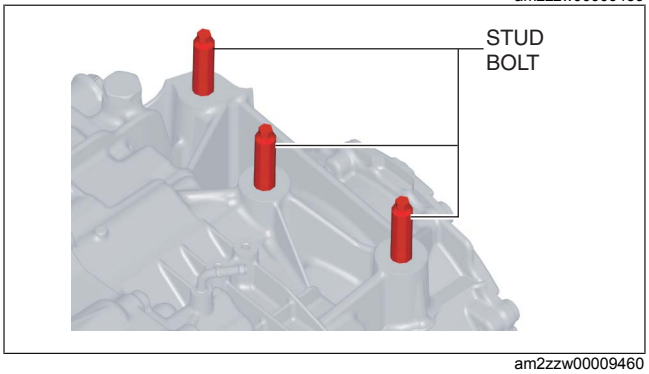


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4. Tighten the stud bolts for the transaxle.

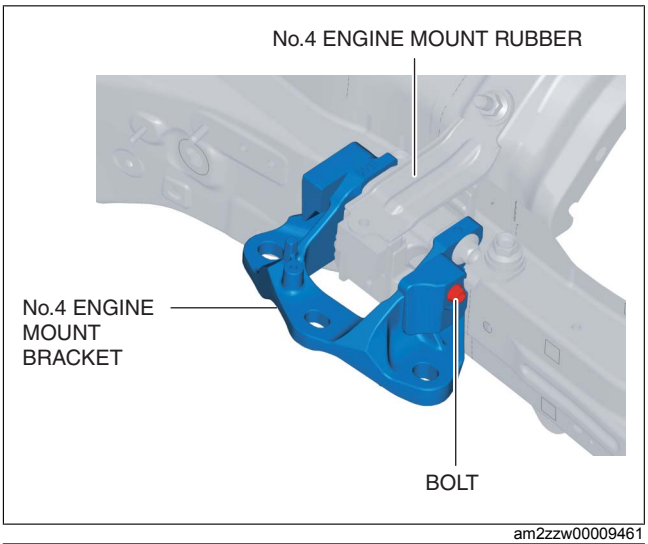
   Note
   - If the No.4 engine mount bracket nut is loosened, tighten the stud bolts for the transaxle because they may be loosened.

   Tightening torque
   15—25 N·m {1.6—2.5 kgf·m, 12—18 ft·lbf}



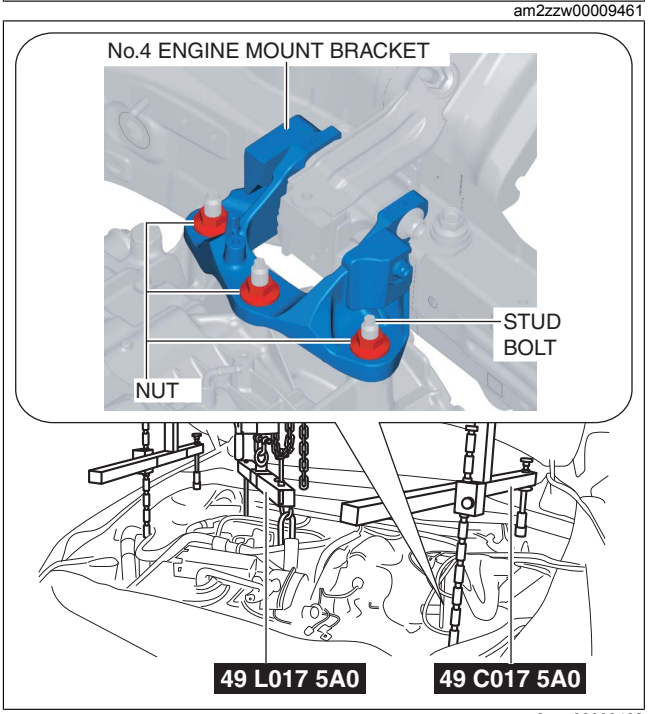
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5. Install the No.4 engine mount bracket to No.4 engine mount rubber, and temporarily tighten the installation bolt.



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6. Lift up the manual transaxle using the **SST** (49 C017 5A0), pass the stud bolt through the No.4 engine mount bracket, and temporarily tighten the No.4 engine mount bracket installation nuts.

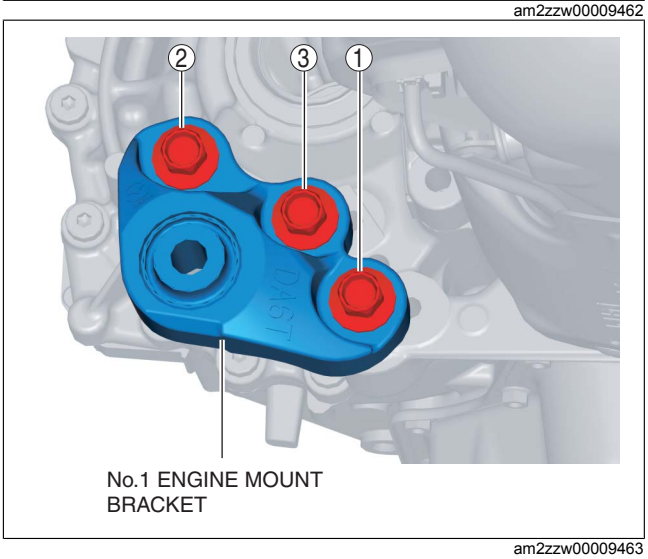


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7. Install the No.1 engine mount bracket, and tighten the installation bolts in the order shown in the figure.

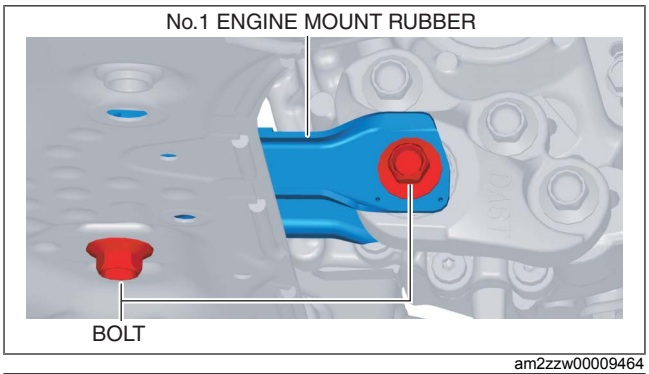
   Tightening torque
   84—98 N·m {8.6—9.9 kgf·m, 62—72 ft·lbf}

8. Install the front crossmember component and No.1 engine mount rubber as a single unit. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



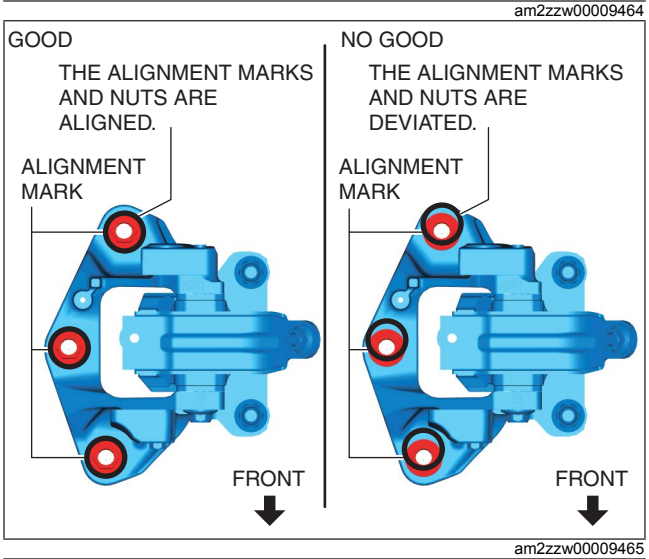
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9. Temporarily tighten the No.1 engine mount rubber installation bolts.



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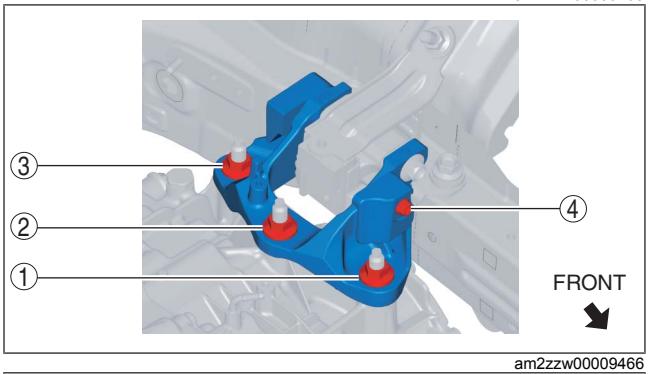
10. Align the alignment marks on the No.4 engine mount bracket and nuts, and temporarily tighten the nuts shown in the figure.



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11. Tighten the No.4 engine mount bracket installation nuts and bolt in the order shown in the figure.

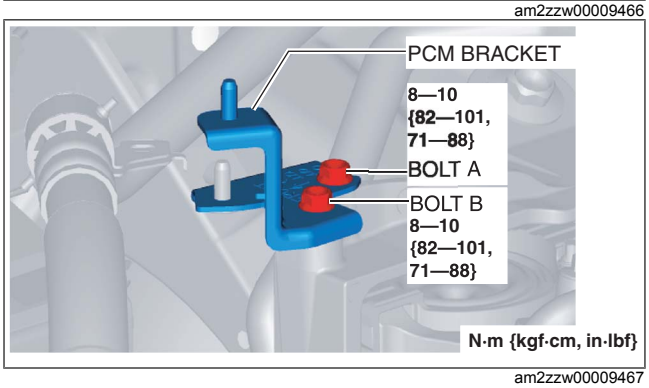
No.	Tightening torque
1, 2, 3	92—116 N·m {9.4—11 kgf·m, 69—85 ft·lbf}
4	81—99 N·m {8.3—10 kgf·m, 60—73 ft·lbf}



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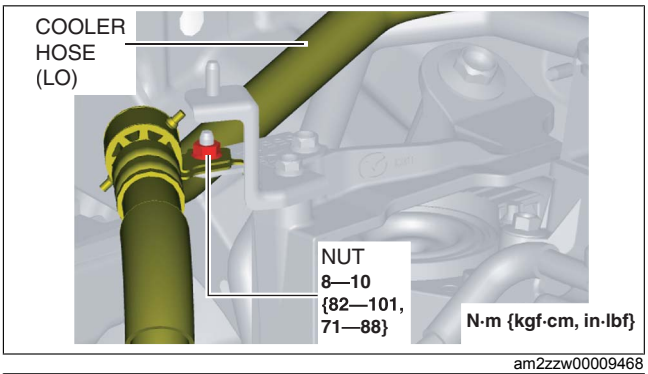
12. Remove the **SST** (49 C017 5A0, 49 L017 5A0).

13. Install the PCM bracket in the following order.
    (1) Tighten bolt A and position the PCM bracket.
    (2) Tighten in the order of bolts B, A.



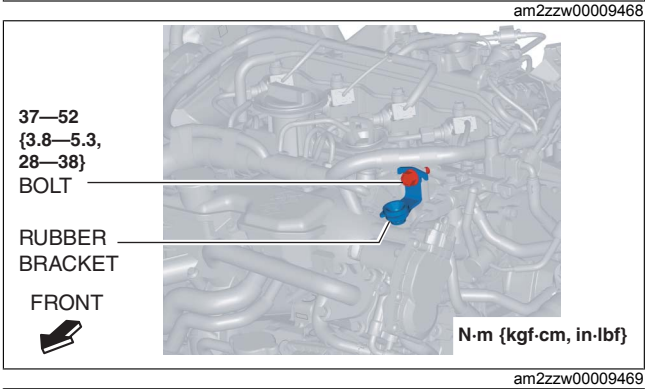
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14. Install the cooler hose (LO).
15. Install the PCM component. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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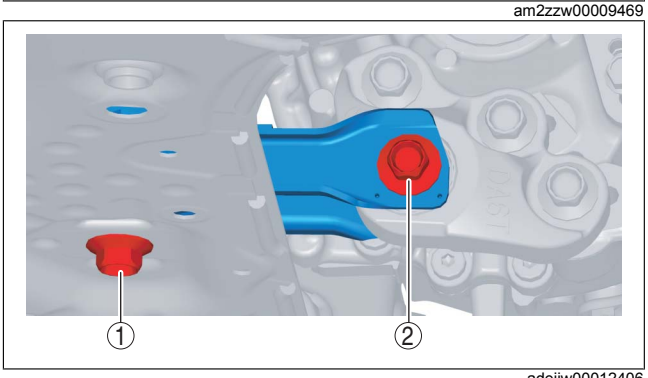
16. Install the rubber bracket.



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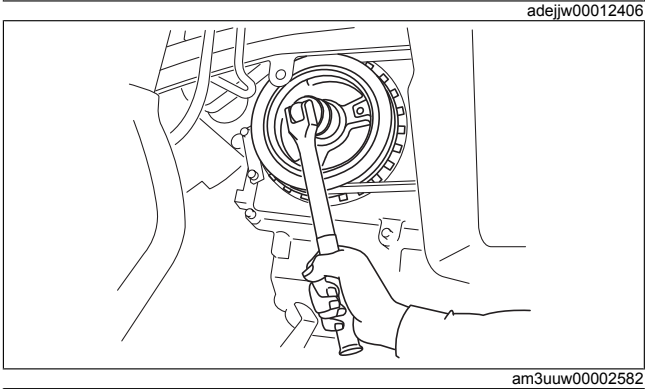
17. Tighten the No.1 engine mount rubber installation bolts in the order shown in the figure.

No.	Tightening torque
1	141—172 N·m {15—17 kgf·m, 104—126 ft·lbf}
2	130—164 N·m {14—16 kgf·m, 96—120 ft·lbf}



adejjw00012406

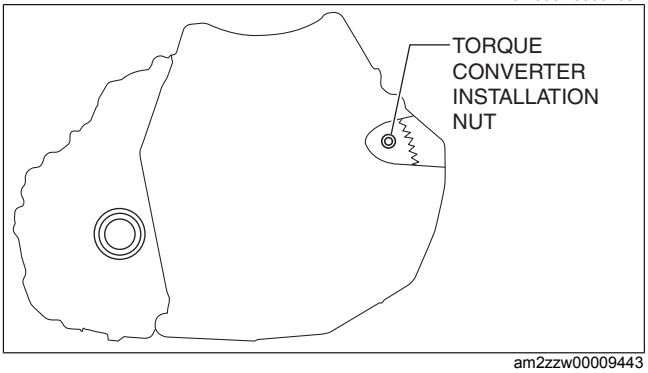
18. Fix the crankshaft pulley to lock the torque converter against rotation.


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19. Tighten the torque converter installation nuts.

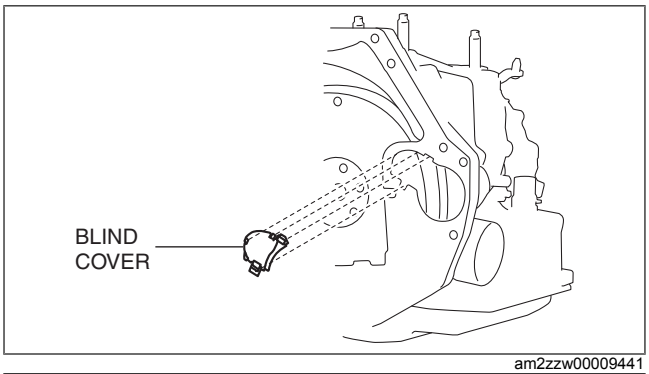
Caution
- **After temporarily tightening the torque converter installation nut uniformly, tighten it to the specified torque.**

Tightening torque
41—54 N·m {4.2—5.5 kgf·m, 31—39 ft·lbf}



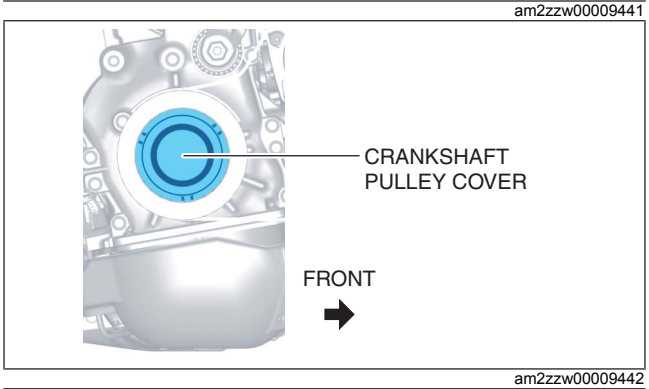
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20. Install the blind cover.
21. Install the starter. (See STARTER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



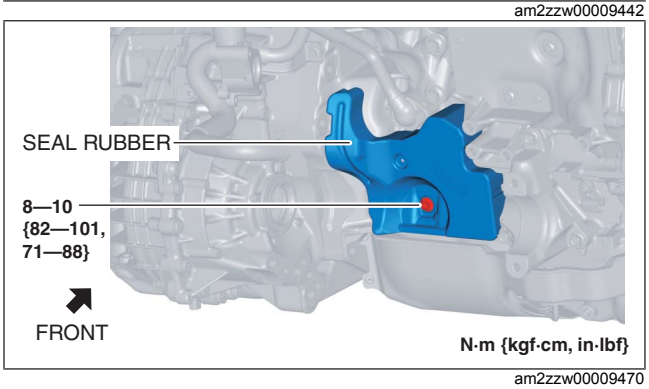
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22. Install the crankshaft pulley cover.
23. Install the electric water pump. (See ELECTRIC WATER PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
24. Install the insulator. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



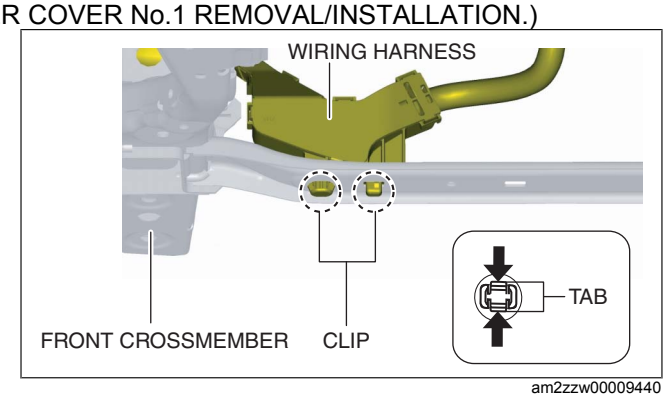
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25. Install the seal rubber.
26. Install the front drive shaft (RH) to the transaxle. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
27. Install the front drive shaft (LH) to the transaxle. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
28. Install the front stabilizer control links to the front stabilizer. (See FRONT STABILIZER INSTALLATION.)
29. Install the front lower arms to the steering knuckles. (See FRONT LOWER ARM REMOVAL/INSTALLATION.)



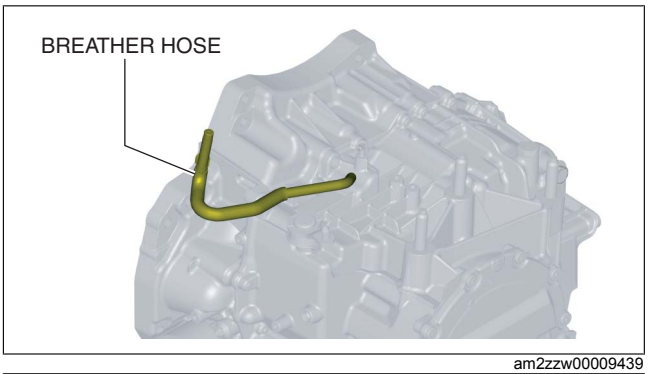
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30. Install the tie-rod ends to the steering knuckles. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
31. Install the clips securing the brake hose to the front shock absorbers. (See BRAKE HOSE (FRONT) REMOVAL/INSTALLATION.)
32. Install the front ABS wheel-speed sensors to the steering knuckles. (See FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION.)
33. Install the front under cover No.1. (See FRONT UNDER COVER No.1 REMOVAL/INSTALLATION.)
34. Connect the wiring harness from the front crossmember. (With i-ELOOP)
35. Install the front tires. (See GENERAL PROCEDURES (SUSPENSION).)
36. Install the intermediate shaft to the steering gear and linkage. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
37. Install the joint cover. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)



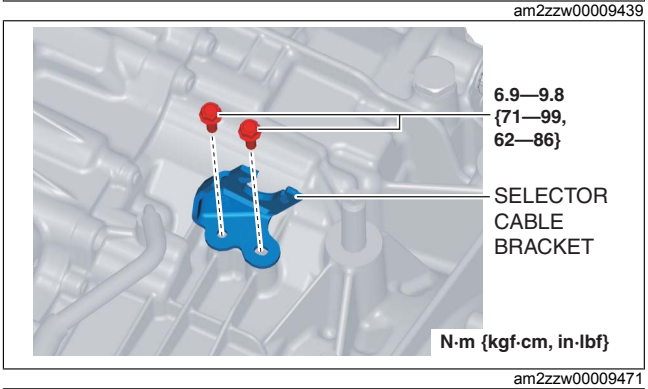
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38. Connect the breather hose to the transaxle.



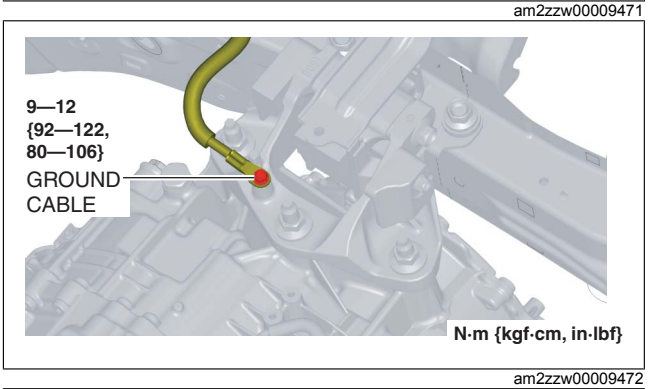
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39. Install the selector cable bracket.
40. Connect the selector cable and clip to the transaxle. (See AUTOMATIC TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION.)



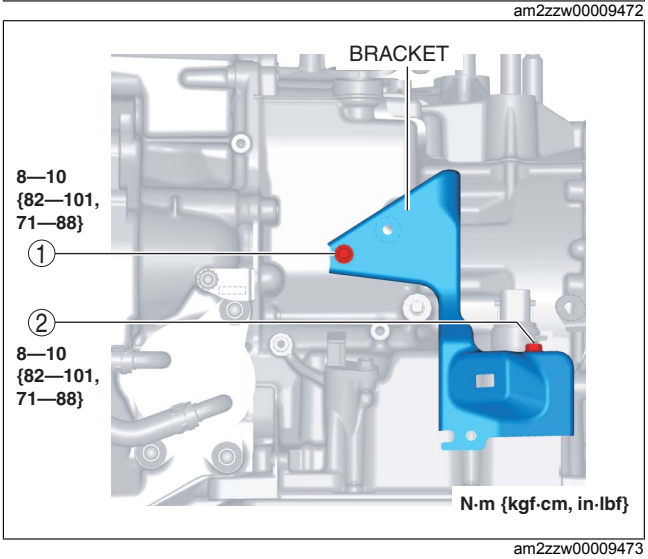
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41. Install the ground cable to the No.4 engine mount bracket.



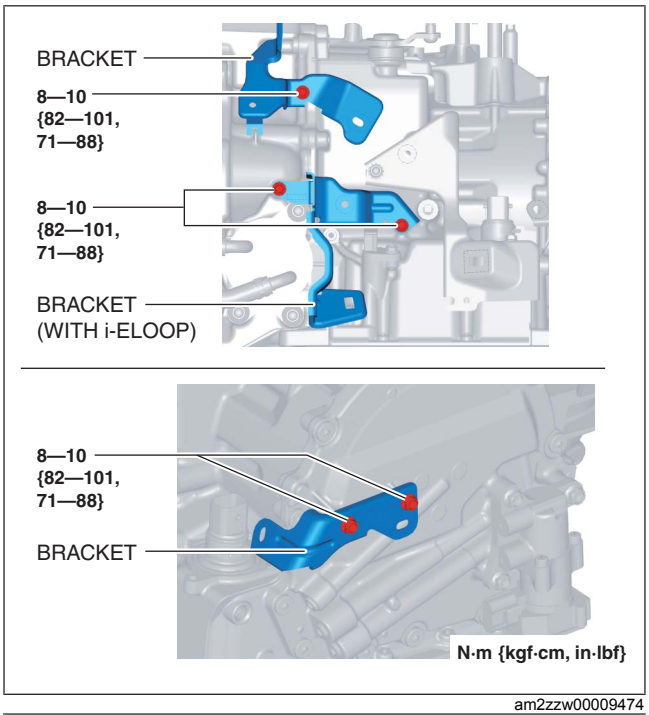
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42. Install the bracket installation bolts in the order shown in the figure.



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43. Install the brackets to the transaxle.

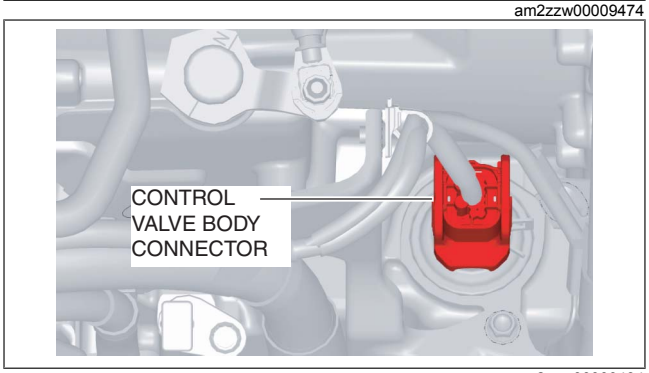


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44. Connect the control valve body connector.

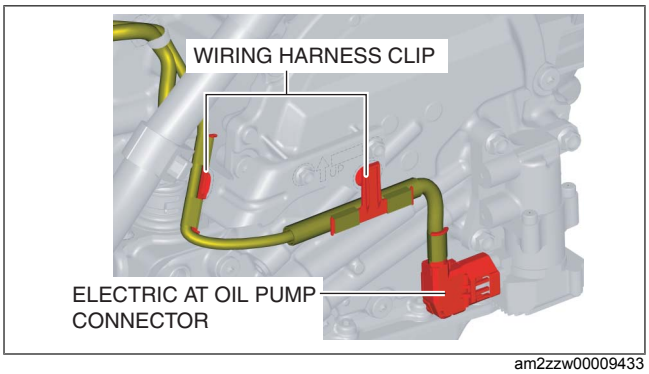
Caution

- **Make sure that your hand does not touch the terminal as the connector terminal could be damaged.**
- **Verify that there is no fluid or foreign matter adhering to the connector before connecting the connector.**
- **Insert the connector straight as the connector terminal could be damaged.**
- **Rotate the connector lever until a click is heard.**



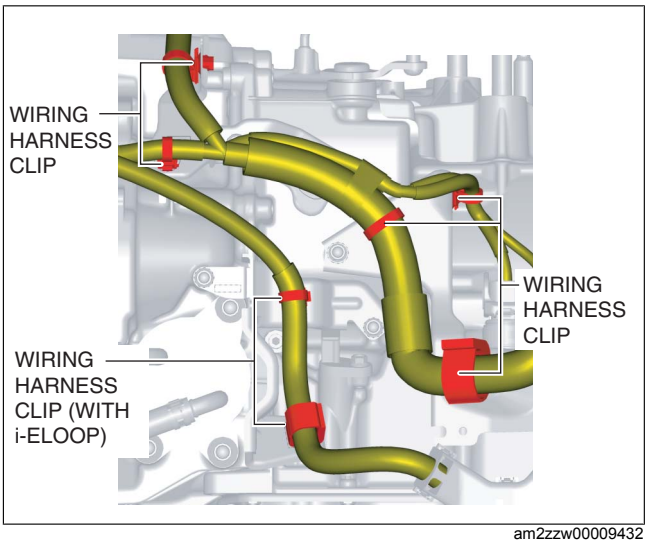
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45. Connect the electric AT oil pump connector and wiring harness clip from the transaxle.



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46. Connect the wiring harness clip to the transaxle.
47. Install the DC-DC converter (i-ELOOP). (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
48. Connect the service plug. (With i-ELOOP) (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].)
49. Install the splash shield (LH). (See SPLASH SHIELD REMOVAL/INSTALLATION.)
50. Install the battery and battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
51. Install the air cleaner, air hose and fresh air duct as a single unit. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
52. Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
53. Add the engine coolant and water-cooled charge air cooler coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].) (See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
54. Add the ATF. (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [EW6A-EL].)
55. Perform the “TCM configuration” (Automatic Transaxle Replacement). (See TCM CONFIGURATION [EW6A-EL].)
56. Perform the “Initial Learning” (automatic transaxle Replacement). (See INITIAL LEARNING [EW6A-EL].)
57. Perform the “Mechanical System Test”. (See MECHANICAL SYSTEM TEST [EW6A-EL].)
58. Install the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)



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